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Studierichting: Informatica
Oefeningenreeks 4A nr. 49.6

$$I = \int \frac{dx}{2\sec x + 3\tan x}$$
$$= \int \frac{dx}{2\frac{1}{\cos x} + 3\frac{\sin x}{\cos x}}$$
$$= \int \frac{dx}{\frac{2+3\sin x}{\cos x}}$$
$$= \int \frac{\cos x}{2+3\sin x} dx$$

Stel $t = 2 + 3\sin x$

$$\Rightarrow dt = 3\cos x \cdot dx$$
$$\Leftrightarrow dx = \frac{dt}{3\cos x}$$
$$\Rightarrow I = \int \frac{dt}{3t}$$
$$= \frac{1}{3} \cdot \ln|t| + c$$
$$= \frac{1}{3} \cdot \ln|2 + 3\sin x| + c$$

References